Exam Center: Western International School

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Mock Junior High School National Exam

Date of Examination: .....

Subject: Physics grade 9

Duration: 45 mins

Scores: 35 marks

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1

Questions and Exercises

I. Multiple choice questions

1. The principle of moment states that:

A. For a body to be in rotational equilibrium, the sum of clockwise moment must equal to the sum of anti-clockwise torques about the same point.

B. For a body to be in rotational equilibrium, the sum of clockwise force must equal to the sum of anti-clockwise torques about the same point.

C. For a body to be in rotational equilibrium, the sum of clockwise moment to the sum of anti-clockwise force about the same point.

D. For a body to be in rotational equilibrium, the sum of clockwise direction to the sum of anti-clockwise force about the same point.

2. The formula of moment is defined as

A. $M = F \times d$ B. $M = F/d$ C. $M = d/F$ D. $M = F + d$

3. Mechanical advantage of simple machine states that:

A. is the number of load a machine multiplies the input (effort).

B. is the number of forces a machine multiplies the effort (input force).

C. is the number of times a machine multiplies the effort (input force).

D. is the number of efforts a machine multiplies the load (output force).

4. A transformer is a step-down transformer when……

A. $\frac{N_S}{N_P} = 0$ B. $\frac{N_S}{N_P} > 1$ C. $\frac{N_S}{N_P} = 1$ D. $\frac{N_S}{N_P} < 1$

5. What is the element resistance of a heating rated as 1200W,240V?

A. $5\,\Omega$ B. $48\,\Omega$ C. $12\,\Omega$ D. $120\,\Omega$

II. Complete each statement with correct terminology or phrase as following

1. The length from the pivot to the line of action of force is called ……

2. A flat supporting surface tilted at an angle with one end higher than the other used as an aid for raising or lowering a load is called……

III. Match the words in part A with the correct definition or corresponding phrase in part B as following

Part A	Part B	Answer
❶ Moment of force ❷ Force ❸ Perpendicular distance	a. Meter (m) b. Newtons meter (N.m) c. Newton (N)	❶ → ❷ → ❸ →

IV. Work problems

1. (5Marks) Calculate the resultant force of two forces ($F_1 = 4$ N, $F_2 = 12$ N) acting on a body. The two forces have the same original point and perpendicular to each other.

2. (5Marks) A tilted road is $50\ \mathrm{m}$ long and $5\ \mathrm{m}$ high above the ground level. Calculate the work done and the force used to move a 100kg object to the top of road (Neglecting friction and $g = 10\mathrm{N/kg}$).

3. (5Marks) A $1000\ \mathrm{W}$ electric heater is used to boil $2\ \mathrm{kg}$ of water from $20\ \mathrm{degrees}$ until boil. Assume that there is no energy lost and specific capacity of water is $4200\ \mathrm{J/kg^0C}$.

A. Calculate heat energy absorbed by water.

B. Calculate the time taken to boil water

C. Calculate the cost per day if we boil $4\ \mathrm{times}$ a day and the price per $1\ \mathrm{kWh}$ is $720\ \mathrm{Riels}$.

4. (5Marks) A transformer has $600\ \mathrm{turns}$ at secondary coil and 200turns at the primary coil.

A. Is it the step up or step-down transformer?

B. Find the primary voltage if the secondary voltage is $240\mathrm{V}$.

C. Find the current in secondary coil if the primary current is $2\ \mathrm{A}$.

5. (7Marks) A $10\ \mathrm{cm}$ high bright object is placed at $25\ \mathrm{cm}$ in front of a convex lens who focal length $15\ \mathrm{cm}$. Determine the image distance.

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2

Questions and Exercises

I. Multiple choice questions

1. The earth wire is used for

a. Preventing getting electric shock. b. Preventing overload circuit or appliances.

c. Preventing exceed electric current passes through the electrical circuit and appliances.

d. Preventing the change in potential difference or voltage across each component.

2. Which of the following is the formula of heat energy?

a. $E = RI^2t$ b. $E = VIt$ c. $E = Pt$ d. $E = mc\Delta T$

II. Complete each statement with the correct terminology or corresponding phrase as following

1. Switch and fuses are connected to …… wire.

2. The longer arm length of inclined plane, the …… of mechanical advantage.

3. In the magnetic field region, conductor carrying electric current is exerted …… and move.

III. Questions

1. What is lens? How many types of lenses are there? What are they?

2. What is lever? How many classes of lever? What are they?

IV. Match the words in part A with the correct definition or corresponding phrase in part B as following

Part A	Part B	Answer
❶ Electric power ❷ Ratio between number of turns and voltage ❸ Mechanical advantage of belt system ❹ Mechanical advantage of gear	a. $MA = N_R/N_E$ b. $P = VI$ c. $MA = F_R/F_E$ d. $V_S/V_P = N_S/N_P$	❶ → ❷ → ❸ → ❹ →

V. Work problems

1. A $12\ \text{cm}$ high object is placed at $30\ \text{cm}$ in front of a convex lens whose focal length is $20\ \text{cm}$.

A. Calculate the image location

B. Calculate the image size

2. An electric lamp having a power of $88\ \text{W}$ is connected a voltage of $220\ \text{V}$.

A. Find the electric current flowing through the lamp.

B. Find the resistance of the lamp

c. The lamp is used $7\ \text{hours}$ per day, calculate the cost of using this lamp for 30days if $1\ \text{kWh}$ costs 720 riels.

3. A transformer has $600\ \text{turns}$ at secondary coil and $200\ \text{turns}$ at the primary coil.

A. Is it the step up or step-down transformer?

B. Find the primary voltage if the secondary voltage is $240\ \text{V}$.

C. Find the current in secondary coil if the primary current is $2\ \text{A}$.

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3

Questions and Exercises

I. Multiple choose questions

1. The three types of beams of light are

a. Diverging, parallel and converging
b. Diverging, concave and converging
c. Concave, convex and straight
d. Converging, concave and straight

2. The certain value of speed of light travel in space approximately

a. $3.00\times10^{8}\ \mathrm{m/s}$ **b.** $225\times10^{3}\ \mathrm{km/s}$ **c.** $2.00\times10^{5}\ \mathrm{km/s}$ **d.** $3.00\times10^{8}\ \mathrm{km/s}$

3. Which of the following is a step-up transformer?

a. $N_S/V_P > 1$ **b.** $N_S/V_P = 1$ **c.** $N_S/V_P < 1$ **d.** $N_S/V_P \geq 1$

4. Which of the following materials used to differentiate white color?

a. Plane mirror **b.** Converging lens **c.** Diverging lens **d.** Prism

II. Questions

1. What is focal length of optical lens?

2. What is mechanical advantage of lever?

III. Match the words in part A with the correct definition or corresponding phrase in part B as following

Part A	Part B	Answer
❶ Refractive index ❷ Refractive index of water ❸ Refractive index of glass ❹ Refractive index of diamond	a. 2.5 b. $n = c/v$ c. 1.33 d. 1.5	❶ → ❷ → ❸ → ❹ →

IV. Work problems

1. We want to lift a 200 N object using a 1.5 m long metal rod to be a lever. The effort is applied at other ends. Calculate the effort force if the fulcrum is at 30 cm from load.

2. A 1000W electric heater is used to boil 2 kg of water from 20 degrees until boil. Assume that there is no energy lost and specific capacity of water is $4200\mathrm{J/kg^{\circ}C}$.

A. Calculate heat energy absorbed by water

B. Calculate the time taken to boil water

C. Calculate the cost per day if we boil 4 times a day and the price per 1 kWh is 720 Riels.

3. A 20 cm high object is placed at 40 cm in front of a converging lens. The image appears at 10 cm behind

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Questions and Exercises

I. Choose the correct answers

1. (2Marks) Mechanical advantage of a machine is defined as:

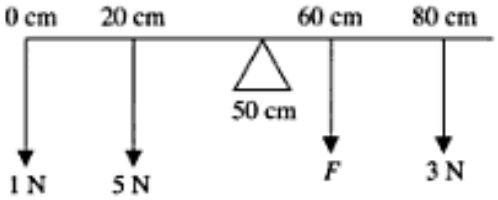
a. $MA = \frac{\text{Effort}}{\text{Load}}$ **b.** $MA = \frac{\text{Force}}{\text{Load}}$ **c.** $MA = \frac{\text{Weight}}{\text{Load}}$ **d.** $MA = \frac{\text{Load}}{\text{Effort}}$

2. (2Marks) A body can have:

a. two centers of gravity **b.** 3 centers of gravity

c. more centers of gravity **d.** Only one center of gravity

3. (2Marks) A long uniform beam is being balanced as shown Calculate the force F.



a. $F = 2\,\text{N}$ **b.** $F = 3\,\text{N}$ **c.** $F = 7\,\text{N}$ **d.** $F = 11\,\text{N}$

4. (2Marks) When you stand $5\ \text{m}$ in front of a plane mirror, your image is located,

a. $5\ \text{m}$ in front of the mirror. **b.** $5\ \text{m}$ behind the mirror.

c. less than $5\ \text{m}$ behind the mirror. **d.** more than $5\ \text{m}$ behind the mirror.

II. Complete each statement with the correct definition or corresponding phrase as following

1. The material which allowed light, but not detail shape is called .....

2. The material which does not given out light is called .....

3. An object is ..... If you can see through it clearly.

4. A region where the very dark part of a shadow is called a .....

III. Match the words in part A with the correct definition or corresponding phrase in part B as following

Part A	Part B	Answer
❶ Voltage	**a.** Joules (J)	❶ →
❷ Electric energy	**b.** Amperes (A)	❷ →
❸ Electric current	**c.** Volts (V)	❸ →
❹ Resistance	**d.** Ohms (Ω)	❹ →

IV. Work problems

Teacher's name: Moeun Sokmean **Tel: 010/015 34 96 34/085 808920**

1. When an electric kettle is connected to a $220\ \text{V}$ mains supply, a current of $4.0\ \text{A}$ passes through it.

(a) (3Marks) When the kettle is operating normally, what current is flowing in

(i) the live wire, _____ A

(ii) the neutral wire, _____ A

(iii) the earth wire, _____ A

(b) (1Marks) Calculate the power supplied to the kettle.

(c) (1Marks) Suggest a rating that could be used with this kettle.
(Available: Fuses of $2\ \text{A}$, $5\ \text{A}$, $10\ \text{A}$, $13\ \text{A}$ ratings)

(d) (1Marks) Calculate the energy in kWh consumed by the kettle when it is switched on for $10\ \text{hours}$.

(e) (1Marks) The metal casing of the electric kettle is connected to the earth.
Give one reason for this.

2. (5 marks) An electric kettle is used to boil $2\ \text{kg}$ of water from $50\ °\text{C}$. The kettle has a power rating of $1\ \text{kW}$ and an efficiency of 80%. How long does the water take to be boiled, provided that the specific heat capacity of water is $4200\ \text{J/kg.°C}$?

3. (10 marks) A transformer has 800 primary turns and 2000 secondary turns. If the primary voltage is $160\ \text{V}$, determine the secondary voltage assuming an ideal transformer.

4. (5 marks) A concave lens has the focal lenth of $10\ \text{cm}$ and the object is palce at $12\ \text{cm}$. Find the image distance.

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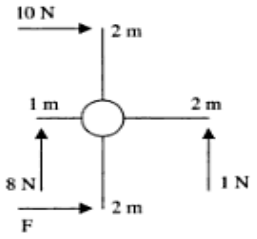
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Questions and Exercises

I. Choose the correct answers

1. (2 marks) A windmill is pushed by four external forces as shown. Calculate force F required to make the windmill in equilibrium.



a. $F = -10$ N **b.** $F = 10$ N

c. $F = 13$ N **d.** $F = 17$ N

2. (2 marks) The figures below show the cross section of four solid objects. Which object is the most stable?



3. (2 marks) To determine the direction of the magnetic field produced by a straight current-carried wire, a right-hand rule is used where the direction the thumb points toward represents

a. the north pole. **b.** the direction of current flows.

c. the south pole. **d.** None of the above.

4. (2 marks) For concave curved mirror, the incident light that goes through the focal point will … off the mirror parallel to the principal axis.

a. refract **b.** reflect

c. incident **d.** diverge

II. (5marks) Match the column A with the column B

	A.Words		B.Meaning
1	Earth wire	A	It has the voltage about 220 Volts to your house and has the color of red or brown.

2	Live wire	B	A wire that is connected from the metal body of the appliances to Earth and, for normal function, it carries no electric current.
3	circuit breaker	C	a lever with wheel which can rotate about a fixed axis.
4	wheel and axle	D	An electromagnetic device that breaks a circuit when the current goes above a certain value.
5	gear	E	a wheel with teeth on its outer edge.

1	2	3	4	5

III. Problems

1. (4 marks) An electric kettle takes in 2000 J of energy per second. It transfers 1500 J as useful heat energy and 500 J is wasted as sound energy. What is the efficiency of the kettle?

2. (3 marks) A cylinder of a wheel and axle having a diameter of 40 cm is used to lift a crate of 150 N. The force applied to the handle is only 100 N, Calculate the length of handle.

3. (8 marks) A water heater has a volume of 10 litres and is fitted with a 2-kW immersion heater. Calculate the time taken to raise the temperature of water from 40 °C to 740 °C. The water heater is 80 percent efficient. The specific heat capacity of water is 4200 J/kg.°C.

4. (2 marks) Calculate the speed of light in a plexiglass if the refractive index of plexiglass is 1.51.

5. (5 marks) A 2.25-cm-tall object is 8.5 cm to the left of a convex lens of 5.5-cm focal length. Find the image position and height.

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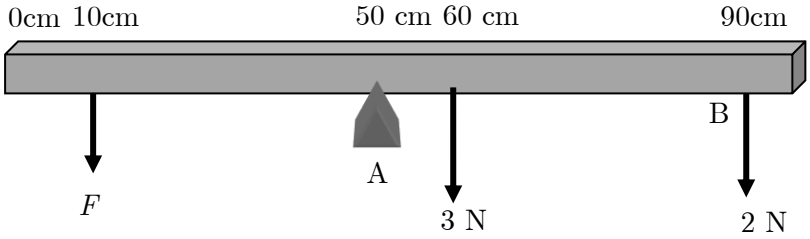
Questions and Exercises

I. Choose the correct answers (8Marks)

1. A fan, with a resistance of 100 Ω, is powered by a 240 V socket. Suggest a suitable rating for a fuse to protect the fan from surge.

a. 1 A **b.** 2 A **c.** 3 A **d.** 5 A

2. A uniform beam of 1m is being balanced as shown below. What is the moment of the 3 N force about the pivot (point B).



a. 10 Nm **b.** 30 Nm **c.** 60 Nm **d.** 90 Nm

3. For converging lenses (convex lenses), the incident light that is parallel to the principle axis will … through the focal point.

a. refract **b.** reflect
c. incident **d.** diverge

4. Which is the most appropriate fuse rating for a 240V, 1.1kW heater?

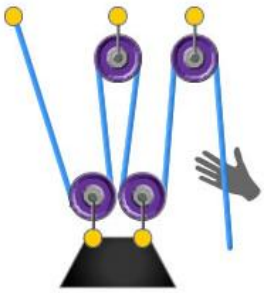
a. 1 A **b.** 5 A **c.** 13 A **d.** 3 A

II. Match the column A with the column B (4Marks)

A	B	Answer
1.Electric heater	a. has the highest potential	❶ →
2. Live wire.	b. is any process in which electrical energy is converted to heat.	❷ →
3. Optical lens	c. A device converts electric energy to heat energy	❸ →
4. Electric heating	d. is an optical device that is used to bend light in a specific way.	❹ →

III. Problems

1. (4 marks) For the pulley System below, find the MA of this pulley system.



2. (3 marks) A uniform meter stick weighs $1.4\ \mathrm{N}$, and is supported by a pivot at the $65\ \mathrm{cm}$ mark. Where must the $0.85\ \mathrm{N}$ force be suspended to balance the stick?

3. (3 marks) A driver gear has 12 teeth and revolves at $100\ \mathrm{turns}$ in 1 minute. The driven gear has 36 teeth. What is the number of turns of the driven gear in 1 minute?

4. (5 marks) A transformer primary winding connected across a $415\ \mathrm{V}$ supply has $750\ \mathrm{turns}$. Determine how many turns must be wound on the secondary side if an output of $1660\ \mathrm{V}$ is required.

5. (5 marks) A 10-cm high object is placed $30\ \mathrm{cm}$ from a 20-cm focal length converging lens. Determine the image distance, the magnification of the image, the image height and the properties of the image.

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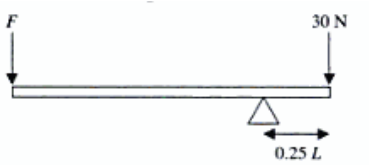
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Questions and Exercises

I. Choose the correct answers

1. (3Marks) Two downward forces, F and $30\ \text{N}$, act on a light uniform plank of length L at the two ends as shown. Calculate the force F to make the plank equilibrium.



a. 10 N **b.** 30N **c.** 40N **d.** 120N

2. (2Marks) When a light ray travels from air into water, it

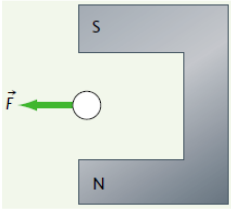
a. reflects and refracts. **b.** converges and diverges.

c. reflects only at the point of incident. **d.** refracts only.

3. (3Marks) Fill in the table below:

Output Force (N)	Input Force (N)	Mechanical Advantage
200		5
	100	4
500	250	

4. (2Marks) Choose the direction of current for the given magnetic force and Field below.



a. To the left **b.** into page **c.** Out of page **d.** upwards

II. Problems

1. (6 Marks) A screw-type pump is used to raise $200\ \text{kg}$ of water a distance of $3\ \text{m}$. The handle of the pump is turned with an effort force of $40\ \text{N}$ a total of 1200 times tracing a circle of $20\ \text{cm}$ circumference each time.

(a) Calculate the work output.

(b) Calculate the work input.

(c) Calculate the efficiency.

2. (6 Marks) A $6\ \mathrm{m}$ light beam is supported at the ends A and B. The weight of $50\ \mathrm{kN}$ is placed at $2.5\ \mathrm{m}$ from A. Find the downward forces exert on the supports A and B.

3. (6 Marks) Where must an object be placed to form an image $30\ \mathrm{cm}$ from a diverging lens with a focal length of $40\ \mathrm{cm}$? Determine the magnification of the image.

4. (7 Marks) A consumer requires an immersion heater for a tank containing $150\ \mathrm{liters}$ of Water. The water is to be heated from $10\ °\mathrm{C}$ to $60\ °\mathrm{C}$ in $3\ \mathrm{hours}$. Calculate the element size (in kilowatts) if the heating system is 90 percent efficient. The specific heat capacity of water is $4200\ \mathrm{J/kg.°C}$.

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Questions and Exercises

I. Choose the correct answers

1. Transformers are used to increase/decrease voltage. If we desire a secondary voltage to be smaller than a primary voltage, then

a. the secondary coil needs to has less turns than the primary coils.

b. the secondary coil needs to has more turns than the primary coils.

c. the number of turns in both coils have to be equal.

d. the ratio between the number of turns in both coils must be smaller than zero.

2. Which rule gives the direction of the magnetic field lines around a current carrying wire?

a. Right-hand grip rule **b.** Left-hand grip rule

c. Fleming's left-hand rule **d.** Right-hand rule

3. Which of the following is **not** a way to increase the speed at which a d.c motor turns?

a. Increasing the number of turns on the coil. **b.** Increase the magnetic field.

c. Move the potential difference. **d.** Decrease the current.

II. Complete each statement with the correct terminology and phrase in the blank as following.

A. Concave mirror	**B.** Inverted	**C.** Convex mirror	**D.** Penumbra	**E.** Umbra
F. Plane mirror	**G.** Upright	**H.** Translucent	**I.** Transparent	**J.** Opaque

i). A flat, smooth surface that reflects light and forms an image is called a ……

ii). An object does not allow light to travel through it is called a …….

iii). Upside down (describing an image of an object) is called the ……

iv). The material which allowed light, but not detail shapes is called ……

v). A …… is a reflecting surface that curves outward like the back of a spoon.

III. Questions

1. List the factors affecting the strength of an electromagnet?

2. What is the transformer? What is the function of transformer?

IV. Match the words in part A with the correct definition or corresponding phrase in part B as following

Part A	Part B	Answer
❶ $M = F \times d$	**a.** Image size	❶ →
❷ $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$	**b.** Moment	❷ →
❸ $\frac{h_i}{h_o} = -\frac{d_i}{d_o}$	**c.** Focal length	❸ →

V. Work problems

1. The convex lens of a copy machine has a focal length of $25.0\ \mathrm{cm}$. A letter to be copied is placed $40.0\ \mathrm{cm}$ from the lens.

a. How far from the lens is the copy paper?

b. How much larger will the copy be?

2. Calculate the refraction index for a substance if the speed of light in that medium is

a. $v = 2.1 \times 10^8\ \mathrm{m/s}$ **b.** $v = 1.5 \times 10^8\ \mathrm{m/s}$ **c.** $v = 0.79\,c$

3. A concave mirror has a radius of $40.0\ \mathrm{cm}$. A 4.0-cm-tall object is $30.0\ \mathrm{cm}$ from the mirror. What is the image position and image height? Is image inverted or upright? Draw rays on the diagram.

4. A convex security mirror in a warehouse has a 0.80-m focal length. A 6.0-m-tall forklift is $10.0\ \mathrm{m}$ from the mirror. What are the image position and image height? Is image inverted or upright? Draw rays on the diagram